

Sec. 4.1. Windows Command Line

Windows Command Line

Task 1 - Introduction

`cmd.exe` is the Windows Command Prompt executable, also known as the **Windows Command Processor**. It is a command-line interpreter that allows users and administrators to execute commands, run scripts, manage files, and perform system administration tasks.

From a cybersecurity perspective, `cmd.exe` is important because attackers commonly use it after gaining access to a system to execute commands, run malicious scripts, or perform reconnaissance. During investigations, security analysts review process creation logs, such as **Windows Event ID 4688** or **Sysmon Event ID 1**, to identify suspicious `cmd.exe` activity, including the executed command, parent process, user account, and command-line arguments.

Example: `cmd.exe /c whoami`

This command runs `whoami` through Command Prompt to identify the current user.

Answer the questions below

What is the default command line interpreter in the Windows environment?
`cmd.exe`

Task 2 - Basic System Information

the command `set` to check your path from the command line. The terminal output below shows the path where MS Windows will execute commands, as indicated by the line starting with `Path=`.

```
C:\>set
ALLUSERSPROFILE=C:\ProgramData
[...]
```

```
LOGNAME=strategos
NUMBER_OF_PROCESSORS=2
OS=Windows_NT
Path=C:\Windows\system32;C:\Windows;C:\Windows\System32\Wbem;
C:\Windows\System32\WindowsPowerShell\v1.0\;C:\Windows\System
32\OpenSSH\;C:\Windows\system32\config\systemprofile\AppData
\Local\Microsoft\WindowsApps;C:\Users\strategos\AppData\Local
\Microsoft\WindowsApps;
[...]
```

the `ver` command to determine the operating system (OS) version.

```
C:\>ver
Microsoft Windows [Version 10.0.17763.1821]
```

the `systeminfo` command to list various information about the system such as OS information, system details, processor and memory.

```
C:\>systeminfo

Host Name:                WIN-SRV-2019
OS Name:                  Microsoft Windows Server 2019 Data
center
OS Version:               10.0.17763 N/A Build 17763
OS Manufacturer:         Microsoft Corporation
OS Configuration:        Standalone Server
OS Build Type:             Multiprocessor Free
[...]
```

- `help` - Provides help information for a specific command
- `cls` - Clears the Command Prompt screen.

Answer the questions below

| What is the OS version of the Windows VM? 10.0.20348.2655

What is the hostname of the Windows VM? WINSRV2022-CORE

Task 3 - Network Troubleshooting

network information using `ipconfig`.

```
C:\>ipconfig
```

Windows IP Configuration

Ethernet adapter Ethernet:

```
Connection-specific DNS Suffix . : eu-west-1.compute.inte
rnal
Link-local IPv6 Address . . . . . : fe80::90df:4861:ba40:f
2a8%4
IPv4 Address. . . . . : 10.10.230.237
Subnet Mask . . . . . : 255.255.0.0
Default Gateway . . . . . : 10.10.0.1
```

`ipconfig /all` for more information about your network configuration.

```
C:\>ipconfig /all
```

Ethernet adapter Ethernet 3:

```
Connection-specific DNS Suffix . : eu-west-1.compute.inte
rnal
Description . . . . . : Amazon Elastic Network
Adapter
Physical Address. . . . . : 02-B7-DF-1D-0D-99
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::90df:4861:ba40:f
2a8%4(Preferred)
IPv4 Address. . . . . : 10.10.230.237(Preferre
```

```
d)
Subnet Mask . . . . . : 255.255.0.0
Lease Obtained. . . . . : Wednesday, May 1, 2024
2:38:05 PM
Lease Expires . . . . . : Wednesday, May 1, 2024
4:08:07 PM
Default Gateway . . . . . : 10.10.0.1
DHCP Server . . . . . : 10.10.0.1
DHCPv6 IAID . . . . . : 134353458
DHCPv6 Client DUID. . . . . : 00-01-00-01-27-E3-D1-2
B-0E-F8-30-D0-72-3F
DNS Servers . . . . . : 10.0.0.2
NetBIOS over Tcpi. . . . . : Enabled
```

Network Troubleshooting: The command syntax is `ping target_name`. Inspired by ping-pong, we send a specific ICMP packet and listen for a response. If a response is received, we know that we can reach the target and that the target can reach us.

```
C:\>ping example.com
```

```
Pinging example.com [93.184.215.14] with 32 bytes of data:
Reply from 93.184.215.14: bytes=32 time=78ms TTL=52
Reply from 93.184.215.14: bytes=32 time=78ms TTL=52
Reply from 93.184.215.14: bytes=32 time=78ms TTL=52
Reply from 93.184.215.14: bytes=32 time=78ms TTL=52
```

```
Ping statistics for 93.184.215.14:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 78ms, Maximum = 78ms, Average = 78ms
```

`tracert`, which stands for *trace route*. The command `tracert target_name` traces the network route traversed to reach the target.

```
C:\>tracert example.com
```

Tracing route to example.com [93.184.215.14]
over a maximum of 30 hops:

```
  1    59 ms    32 ms    42 ms  ec2-3-248-240-3.eu-west-1.com
pute.amazonaws.com [3.248.240.3]
  2      *      *      *      Request timed out.
  3      *      *      *      Request timed out.
  4      *      *      *      Request timed out.
  5      *      *      *      Request timed out.
  6      *      *      *      Request timed out.
  7      *      *      *      Request timed out.
  8      *      *      *      Request timed out.
  9    <1 ms   13 ms    <1 ms  100.100.2.56
 10    15 ms   11 ms   11 ms  ae-42.a03.londen12.uk.bb.gin.
ntt.net [131.103.117.104]
 11    17 ms   11 ms   12 ms  ae-14.r20.londen12.uk.bb.gin.
ntt.net [129.250.3.248]
 12    81 ms   80 ms   80 ms  ae-7.r20.nwrknj03.us.bb.gin.n
tt.net [129.250.6.147]
 13    83 ms   83 ms   86 ms  ae-0.a02.nycmny17.us.bb.gin.n
tt.net [129.250.3.9]
 14    79 ms   79 ms   96 ms  ce-0-3-0.a02.nycmny17.us.ce.g
in.ntt.net [128.241.1.14]
 15    81 ms   86 ms   79 ms  ae-67.core1.nyd.edgecastcdn.n
et [152.195.68.135]
 16    78 ms   78 ms   78 ms  93.184.215.14
```

Trace complete.

`nslookup`. It looks up a host or domain and returns its IP address. The syntax `nslookup example.com` will look up `example.com` using the default name server;

```

C:\>nslookup example.com
Server:  ip-10-0-0-2.eu-west-1.compute.internal
Address:  10.0.0.2

Non-authoritative answer:
Name:     example.com
Addresses: 2606:2800:21f:cb07:6820:80da:af6b:8b2c
          93.184.215.14

C:\>nslookup example.com 1.1.1.1
Server:  one.one.one.one
Address:  1.1.1.1

Non-authoritative answer:
Name:     example.com
Addresses: 2606:2800:21f:cb07:6820:80da:af6b:8b2c
          93.184.215.14

```

`netstat` This command displays current network connections and listening ports. A basic `netstat` command with no arguments will show you established connections, as shown below.

```

C:\>netstat

Active Connections

    Proto Local Address           Foreign Address         State
    TCP    10.10.230.237:22        ip-10-11-81-126:53486  ESTABLISHED
    ISHED

```

If you are curious about the other options, you can run `netstat -h`, where `-h` displays the help page. We opted for the following options:

- `a` displays all established connections and listening ports

- **b** shows the program associated with each listening port and established connection
- **o** reveals the process ID (PID) associated with the connection
- **n** uses a numerical form for addresses and port numbers

We combine these four options and execute the `netstat -abon` command. The result is quite long, but we display the first few lines in the terminal below.

```
C:\>netstat -abon
```

Active Connections

Proto	Local Address	Foreign Address	State
TCP	0.0.0.0:22	0.0.0.0:0	LISTEN
ING	2116		
	[sshd.exe]		
TCP	0.0.0.0:135	0.0.0.0:0	LISTEN
ING	820		
	RpcSs		
	[svchost.exe]		
	[...]		
TCP	0.0.0.0:49669	0.0.0.0:0	LISTEN
ING	2036		
	[spoolsv.exe]		
TCP	0.0.0.0:49670	0.0.0.0:0	LISTEN
ING	584		
	Can not obtain ownership information		
TCP	0.0.0.0:49686	0.0.0.0:0	LISTEN
ING	592		
	[lsass.exe]		
TCP	10.10.230.237:22	10.11.81.126:53486	ESTABL
ISHED	2116		
	[sshd.exe]		
	[...]		

Answer the questions below

Which command can we use to look up the server's physical address (MAC address)? `ipconfig /all`

What is the name of the service listening on port 135? `RpcSs`

What is the name of the service listening on port 3389? `TermService`

Task 4 - File and Disk Management

You can use `cd` without parameters to display the current drive and directory, *where am I?*

You can view the child directories using `dir`.

```
C:\Users\strategos>cd  
C:\Users\strategos
```

```
C:\Users\strategos>dir  
Volume in drive C has no label.  
Volume Serial Number is A8A4-C362
```

Directory of C:\Users\strategos

05/01/2024	02:40 PM	<DIR>	.
05/01/2024	02:40 PM	<DIR>	..
11/14/2018	06:56 AM	<DIR>	Desktop
05/01/2024	02:40 PM	<DIR>	Documents
09/15/2018	07:19 AM	<DIR>	Downloads
09/15/2018	07:19 AM	<DIR>	Favorites
09/15/2018	07:19 AM	<DIR>	Links
09/15/2018	07:19 AM	<DIR>	Music
09/15/2018	07:19 AM	<DIR>	Pictures
09/15/2018	07:19 AM	<DIR>	Saved Games
09/15/2018	07:19 AM	<DIR>	Videos

```
0 File(s)          0 bytes
11 Dir(s)  14,984,953,856 bytes free
```

Note that you can use the following options with `dir`:

- `dir /a` - Displays hidden and system files as well.
- `dir /s` - Displays files in the current directory and all subdirectories.

You can type `tree` to visually represent the child directories and subdirectories.

```
C:\Users\strategos>tree
Folder PATH listing
Volume serial number is A8A4-C362
C:..
|—Desktop
|—Documents
|—Downloads
|—Favorites
|—Links
|—Music
|—Pictures
|—Saved Games
|—Videos
```

You can change to any directory by using the command `cd target_directory`; this is equivalent to double-clicking the `target_directory` on your desktop. Furthermore, you can use `cd ..` to go up one level.

```
C:\>cd
C:\

C:\>cd Users

C:\Users>cd
C:\Users
```

```
C:\Users>cd ..
```

```
C:\>cd
```

```
C:\
```

To create a directory, use `mkdir directory_name`; `mkdir` stands for *make directory*. To delete a directory, use `rmdir directory_name`; `rmdir` stands for *remove directory*.

```
C:\example>mkdir backup_files
```

```
strategos@WIN-SRV-2019 C:\example>dir
```

```
Directory of C:\example
```

```
05/02/2024  07:36 AM    <DIR>          .
05/02/2024  07:36 AM    <DIR>          ..
05/02/2024  07:36 AM    <DIR>          backup_files
               0 File(s)                0 bytes
               3 Dir(s)  14,984,724,480 bytes free
```

```
C:\example>rmdir backup_files
```

```
C:\example>dir
```

```
Directory of C:\example
```

```
05/02/2024  07:36 AM    <DIR>          .
05/02/2024  07:36 AM    <DIR>          ..
               0 File(s)                0 bytes
               2 Dir(s)  14,984,724,480 bytes free
```

Working With Files

view text files with the command `type`. The `copy` command allows you to copy files from one location to another.

```
C:\example>dir
```

```
Directory of C:\example
```

```
05/02/2024  08:12 AM    <DIR>        .
05/02/2024  08:12 AM    <DIR>        ..
05/02/2024  07:57 AM                17 test.txt
          1 File(s)                17 bytes
          2 Dir(s)  14,983,409,664 bytes free
```

```
C:\example>copy test.txt test2.txt
          1 file(s) copied.
```

```
C:\example>dir
Directory of C:\example

05/02/2024  08:12 AM    <DIR>        .
05/02/2024  08:12 AM    <DIR>        ..
05/02/2024  07:57 AM                17 test.txt
05/02/2024  07:57 AM                17 test2.txt
          2 File(s)                34 bytes
          2 Dir(s)  14,983,409,664 bytes free
```

you can move files using the `move` command.

```
C:\example>dir
Directory of C:\example

05/02/2024  08:12 AM    <DIR>        .
05/02/2024  08:12 AM    <DIR>        ..
05/02/2024  07:57 AM                17 test.txt
05/02/2024  07:57 AM                17 test2.txt
          2 File(s)                34 bytes
          2 Dir(s)  14,983,409,664 bytes free
```

```
C:\example>move test2.txt ..
          1 file(s) moved.
```

```
C:\example>dir
```

Directory of C:\example

```
05/02/2024  08:13 AM    <DIR>          .
05/02/2024  08:13 AM    <DIR>          ..
05/02/2024  07:57 AM                17 test.txt
          1 File(s)                17 bytes
          2 Dir(s)  14,983,409,664 bytes free
```

Finally, we can delete a file using `del` or `erase`.

```
C:\example>dir
```

Directory of C:\example

```
05/02/2024  08:16 AM    <DIR>          .
05/02/2024  08:16 AM    <DIR>          ..
05/02/2024  07:57 AM                17 test.txt
05/02/2024  07:57 AM                17 test2.txt
          2 File(s)                34 bytes
          2 Dir(s)  14,983,409,664 bytes free
```

```
C:\example>erase test2.txt
```

```
C:\example>dir
```

Directory of C:\example

```
05/02/2024  08:16 AM    <DIR>          .
05/02/2024  08:16 AM    <DIR>          ..
05/02/2024  07:57 AM                17 test.txt
          1 File(s)                17 bytes
          2 Dir(s)  14,983,409,664 bytes free
```

We can use the wildcard character `*` to refer to multiple files. For example, `copy *.md C:\Markdown` will copy all files with the extension `md` to the directory `C:\Markdown`.

Answer the questions below

What are the file's contents in C:\Treasure\Hunt? THM{CLI_POWER}

'Task 5 - Task and Process Management

list the running processes using `tasklist`.

```
C:\>tasklist
```

Image Name	PID	Session Name	Session
n# Mem Usage			
System Idle Process	0	Services	
0 8 K			
System	4	Services	
0 88 K			
Registry	84	Services	
0 50,700 K			
smss.exe	276	Services	
0 1,132 K			
csrss.exe	372	Services	
0 5,264 K			
wininit.exe	448	Services	
0 6,892 K			
csrss.exe	456	Console	
1 5,028 K			
winlogon.exe	516	Console	
1 11,144 K			
services.exe	584	Services	
0 7,492 K			
lsass.exe	592	Services	
0 16,108 K			
svchost.exe	704	Services	
0 23,432 K			
fontdrvhost.exe	736	Console	
1 4,256 K			
[...]			

```
C:\>tasklist /FI "imagename eq sshd.exe"
```

Image Name	PID	Session Name	Session
n# Mem Usage			
=====	=====	=====	=====
sshd.exe	2116	Services	
0 6,992 K			
sshd.exe	2712	Services	
0 7,668 K			
sshd.exe	4752	Services	
0 7,372 K			

With the process ID (PID) known, we can terminate any task using `taskkill /PID target_pid`.

Answer the questions below

What command would you use to find the running processes related to notepad.exe? `tasklist /FI "imagename eq notepad.exe"`

What command can you use to kill the process with PID 1516? `taskkill /PID 1516`

Task 6 - Conclusion

- `chkdsk`: checks the file system and disk volumes for errors and bad sectors.
- `driverquery`: displays a list of installed device drivers.
- `sfc /scannow`: scans system files for corruption and repairs them if possible.

It is important to remember all the commands covered in the previous tasks; moreover, it is equally important to know that `/?` can be used with most commands to display a help page.

In this room, we used the command `more` in two ways:

- Display text files: `more file.txt`
- Pipe long output to view it page by page: `some_command | more`

Answer the questions below

The command shutdown /s can shut down a system. What is the command you can use to restart a system? shutdown /r

What command can you use to abort a scheduled system shutdown? shutdown /a